

Stated Preference Validity in the Age of Immersive Media: Evidence on Scope Sensitivity and Hypothetical Bias

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Abstract¹

Immersive virtual reality (VR) has been proposed as a way to help respondents value unfamiliar environmental goods, yet its effects on stated preference validity remain largely untested. Using a preregistered 2×2 experiment (N=246), we compare fully immersive, multisensory VR with high-quality video for Amazon rainforest conservation under real and hypothetical donations, among the first contingent valuation applications of fully immersive VR. Both formats produced scope-sensitive valuations with comparable elasticities. Neither format reduced hypothetical bias, with hypothetical willingness to pay (WTP) approximately five times higher than real donations. Exploratory within-VR analyses show that perceived realism and emotional engagement were associated with higher hypothetical WTP while real donations remained stable, whereas presence was associated with stronger scope sensitivity without increasing hypothetical bias. These findings suggest that immersive technologies reshape stated preference validity profiles rather than uniformly improving valuation quality, and that high-quality video may provide a lower-cost alternative for environmental valuation applications.

Keywords: Stated Preference, Scope Sensitivity, Hypothetical Bias, Virtual Reality, Contingent Valuation

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Word count: 8430

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1.INTRODUCTION

The Amazon rainforest is approaching ecological tipping points beyond which additional disturbances could trigger abrupt and potentially irreversible losses in ecosystem services (Lovejoy and Nobre 2018; Wunderling et al. 2022). Approximately 17% has already been deforested (Gatti et al. 2021), with projections suggesting that up to 47% may face compounding disturbances by 2050 (Flores et al. 2024). As the timeline for global commitment to protect 30% of natural ecosystems by 2030 approaches, accurately valuing the conservation benefits of such ecosystems is critical for guiding policy and resource allocation. For non-local populations, these benefits consist predominantly of non-use values — existence, bequest, and altruistic values — as most individuals will never directly experience the rainforest (Navrud and Strand 2018; Brouwer et al. 2022). Contingent valuation (CV), the primary tool for measuring such non-market benefits, has long relied on textual descriptions and static images to convey environmental goods (OECD 2018).

When unfamiliar ecosystems are conveyed through these conventional formats, respondents may rely on simplified mental representations rather than carefully evaluating environmental scale or economic trade-offs (Kahneman 2003; Lopes and Kipperberg 2020; Carson 1997). From a constructed preferences perspective, stated valuations are formed during the elicitation process using available representational cues, rather than retrieved as fully pre-existing preferences (Payne, Bettman, and Johnson 1992; Lichtenstein and Slovic 2006). Presentation format may therefore play a substantial role in how respondents understand unfamiliar environmental goods and construct valuations for them.

Virtual reality (VR) offers a qualitatively different representational approach by creating immersive, multisensory experiences that enable more direct perceptual engagement with unfamiliar environments (Patterson et al. 2017). The implications of this richer form of representation for valuation validity are particularly relevant to two established challenges in stated preference methods. The first is scope insensitivity. Economic theory predicts that willingness to pay (WTP) should increase with the scale of conservation benefits, such that protecting 50% of the Amazon should be valued more than protecting 20%. Yet respondents do not always distinguish appropriately between conservation scales, particularly for unfamiliar

environmental goods with high non-use values (Burrows et al. 2017; Kahneman, Ritov, and Schkade 1999). The second is hypothetical bias, where respondents systematically overstate WTP in hypothetical scenarios relative to actual payment contexts (OECD 2018; Penn and Hu 2018). Meta-analyses suggest that hypothetical valuations can exceed real payments by factors ranging from 1.3 to 14 depending on the good being valued (Haghani et al. 2021a; Schmidt and Bijmolt 2020; Penn and Hu 2018).

For scope sensitivity, VR's potential may operate primarily through representational and cognitive channels. Immersive environments can enhance spatial perception and understanding of scale (Azarby and Rice 2022; Sun, Wu, and Cai 2019), potentially improving respondents' ability to distinguish between conservation outcomes of different magnitudes. The sense of presence — feeling psychologically “there” in the environment — may further transform abstract conservation scenarios into more concrete and experientially grounded evaluations (Steuer 1992; Slater 2009). By providing more standardised environmental representations, immersive presentation may reduce heterogeneity in how respondents interpret conservation outcomes relative to abstract descriptive formats (Zhu, Guo, and Zhao 2021), potentially facilitating clearer differentiation between conservation scales.

At the same time, VR may also heighten the experiential intensity of valuation scenarios by making them feel vivid, visceral, and psychologically realistic (Slater 2009). This heightened experiential engagement may encourage deeper involvement with hypothetical valuation tasks, particularly when considering commitments involving personal resources such as monetary donations (Lee and Li 2023). However, immersive environments are also known to heighten emotional engagement and feelings of environmental commitment (Breves and Heber 2020), which may in turn inflate stated preferences for morally desirable goods such as environmental conservation, particularly in hypothetical settings where financial constraints remain non-binding (Kahneman and Knetsch 1992). Under this account, VR could widen rather than narrow the gap between stated and actual behaviour. Formats that strengthen comprehension of environmental scale may simultaneously intensify affective engagement with hypothetical scenarios, creating the possibility that representational improvements benefit some aspects of validity while weakening others.

Only one prior study has directly examined VR's effect on hypothetical bias. Fang et al. (2021) found evidence that VR may reduce bias in food choice experiments using a computer-generated grocery store, although the effect was significant only after excluding participants reporting VR-induced simulator sickness. That study involved familiar private goods with well-defined market referents, leaving VR's implications for environmental valuation, where goods may be psychologically distant, generate predominantly non-use values, and lack market analogues, untested. More broadly, no study has simultaneously examined how immersive presentation affects both scope sensitivity and hypothetical bias.

These literatures have developed largely in parallel, with scope sensitivity research focused on presentation quality and construct validity (Whitehead 2016; Burrows et al. 2017) and hypothetical bias research focused on consequentiality mechanisms and criterion validity (Penn and Hu 2018; Penn, Hu, and Ye 2024; Johnston et al. 2017). Studying these dimensions jointly matters because if immersive technologies improve scope differentiation while amplifying hypothetical overstatement, the net effect on welfare estimates used in cost-benefit analysis is ambiguous, and practitioners require guidance on how presentation format choices affect the reliability of stated preference inputs to policy decisions.

This study addresses these gaps using a preregistered 2×2 between-subjects experiment (N = 246) comparing immersive VR and high-quality video presentations for Amazon rainforest conservation. It also provides one of the first CV applications of fully immersive, multisensory VR. Participants were randomly assigned to VR or video conditions and further randomised into real or hypothetical donation conditions, enabling direct measurement of both scope sensitivity and hypothetical bias within the same experimental framework. The VR condition employed 360° footage, interactive navigation, spatial audio, and controlled environmental conditions, while the video condition presented identical informational content in 2D format, isolating the effect of immersive presentation while holding content constant. Both conditions elicited WTP for conservation programmes protecting 20% and 50% of the Amazon, with the order of scope levels randomised within each cell.

The results suggest that immersive technologies influence different dimensions of stated preference validity through distinct experiential pathways. Both VR and video produced scope-sensitive valuations with empirically similar elasticities, indicating that clear visual

communication of environmental scale may be sufficient for eliciting scope-sensitive valuations. However, VR did not reduce hypothetical bias relative to video, and within-VR analyses reveal that emotional engagement and perceived realism were associated with higher hypothetical WTP without proportionate increases in real donations. Together, these findings suggest that immersive technologies reshape the validity profile of stated preferences rather than uniformly improving valuation quality.

The remainder of the paper is organised as follows. Section 2 reviews the literature on scope sensitivity, hypothetical bias, and VR in stated preference methods. Section 3 describes the experimental design and empirical strategy. Section 4 presents the results, including exploratory analyses of experiential moderators. Section 5 discusses implications, limitations, and directions for future research, and Section 6 concludes.

2.BACKGROUND

2.1. Validity Challenges in Stated Preference Methods

Scope sensitivity remains a persistent validity challenge in stated preference valuation, particularly for non-use environmental goods. Burrows et al. (2017) found that 27 of 59 studies examining non-use values failed external scope tests, with WTP failing to increase significantly with the scale of the good being valued, including 7 of 11 forest valuation studies. Several explanations have been proposed, including diminishing marginal utility and warm glow motivations whereby respondents derive utility from the act of contributing rather than from the environmental outcome itself (Rollins and Lyke 1998; Kahneman and Knetsch 1992). Most relevant to the present study is representational inadequacy, which may be especially problematic when respondents have limited familiarity with the environmental good (Heberlein et al. 2005). When unfamiliar environmental goods are conveyed through text and static images, conservation scenarios may become cognitively compressed into relatively undifferentiated categories, weakening scope discrimination (Tversky and Kahneman 1974; Lopes and Kipperberg 2020).

The hypothetical bias literature documents a second, distinct validity concern. A central explanation is the absence of binding economic consequences. When respondents recognise

that their choices carry no real financial commitment, they may engage less carefully with economic trade-offs (Johnston et al. 2017; Zawojka, Bartczak, and Czajkowski 2019). Existing mitigation approaches therefore seek to strengthen consequentiality through cheap talk scripts, solemn oaths, budget reminders, and ex-post certainty corrections (Cummings and Taylor 1999; Haghani et al. 2021b). Recent evidence, however, suggests that such techniques should be applied cautiously, as solemn oaths and certainty adjustments may themselves alter WTP when applied in consequential settings (Chopra and Vossler 2026). Beyond consequentiality, however, hypothetical bias may also reflect affective and moral motivations. Respondents may derive utility from expressing support for conservation itself rather than from the underlying environmental outcomes, particularly in hypothetical settings where financial constraints remain non-binding (Johansson-Stenman and Svedsäter 2012; Shan et al. 2023). This affective channel becomes especially relevant when considering presentation technologies that heighten emotional engagement with environmental scenarios.

2.2. Virtual Reality as a Presentation Technology

The theoretical channels through which immersive presentation may affect scope sensitivity and hypothetical bias, outlined above, have limited empirical support. For scope sensitivity — where immersive presentation may operate through representational and cognitive channels — visual aids such as graphical risk displays have been associated with modest improvements in scale discrimination (Corso, Hammitt, and Graham 2001). In stated preference contexts, immersive environments have been linked to reduced decision error and improved choice consistency relative to conventional formats (Bateman et al. 2009; Birenboim et al. 2019).

In the study most closely related to the present research, Mokas et al. (2021) found that immersive VR increased respondents' stated certainty and reduced error variance in a discrete choice experiment on urban green infrastructure, with VR displaying the lowest error variance among text, video, and VR formats. They also found that presentation format significantly affected WTP estimates, including preference reversals between text and multimedia conditions, although the study did not test scope sensitivity directly. Whether these findings extend to fully immersive VR with multisensory enhancements in a CV setting, and whether these improvements translate into greater scope sensitivity remains untested.

For hypothetical bias, where the predicted direction of VR's effect is theoretically ambiguous, direct evidence is more limited. Fang et al. (2021) conducted the only direct test, using a between-subjects experiment that varied both the elicitation method (hypothetical vs. non-hypothetical) and the presentation format (text, pictures, or a computer-generated VR grocery store) for food choices. In the non-hypothetical treatments, one of the participants' choice sets was randomly selected as binding, requiring actual purchase at the stated price. They found that VR produced the smallest hypothetical bias among all hypothetical conditions, although the differences across hypothetical formats were not statistically significant in the full sample. After excluding participants reporting VR-induced simulator sickness, the difference between the hypothetical VR treatment and the non-hypothetical benchmark was no longer statistically significant for one product category.

Several features of their design limit generalisability to the present context. The study involved familiar private goods with well-defined market referents, used computer-generated rather than 360° filmed environments, and employed a choice experiment rather than a CV framework. For unfamiliar environmental goods characterised by arguably greater psychological distance and generating predominantly non-use values, the mechanisms through which immersive presentation affects the hypothetical–real gap may differ substantially. More broadly, studies examining VR and pro-environmental behaviour report heterogeneous effects across contexts and outcome measures, with immersive experiences often increasing stated intentions but producing inconsistent effects on actual behaviour (Nelson, Anggraini, and Schlüter 2020; Banerjee and Ferreira 2024).

2.3. Empirical Predictions

The preregistered analysis examines both scope sensitivity and hypothetical bias. First, the study tests whether WTP varies with conservation scope and whether the degree of scope responsiveness differs between VR and video. If immersive environments improve respondents' comprehension of environmental scale, VR may strengthen scope sensitivity relative to video.

Second, the study examines whether hypothetical WTP differs from real donation behaviour and whether this gap varies across presentation formats. As discussed above, the predicted direction of VR's effect on hypothetical bias is theoretically ambiguous. If immersive

presentation makes hypothetical scenarios feel more psychologically consequential, VR may narrow the hypothetical–real gap. If instead VR primarily amplifies affective responses to morally desirable goods, it may widen this gap.

Third, exploratory analyses investigate heterogeneity within VR responses by examining whether experiential features — presence, perceived realism, and emotional engagement — moderate scope sensitivity and hypothetical bias differently. These analyses are intended to provide insight into potential experiential pathways associated with valuation behaviour within VR. As experiential measures are collected post-treatment, the findings are interpreted as associations rather than causal mediation effects.

3. METHODS

The study was preregistered and approved by the Research Ethics Review Board of the London School of Economics and Political Science. All procedures adhered to the Board’s ethical guidelines and regulations.²

3.1. Virtual Reality and Video Design

The core stimulus was developed by combining selected excerpts from multiple third-party immersive 360° audiovisual sources depicting the Amazon rainforest with researcher-created visual and interactive elements.³ The audiovisual components were assembled in Final Cut Pro, with selected material upscaled from 2K to 4K resolution using Topaz Labs to enhance visual clarity. The production was subsequently assembled in 3DVista Pro and deployed to an offline server, incorporating spatial audio and interactive hotspots containing facts about the Amazon ecosystem.

The resulting stimulus followed a nine-minute structured narrative, a duration chosen to allow sufficient spatial immersion while minimising motion sickness risks that increase beyond ten minutes (Zhang et al. 2018; Chang, Kim, and Yoo 2020). Using a Meta Quest 3 headset, participants were presented with diverse Amazon ecosystems featuring wildlife encounters, indigenous community interactions, river journeys, and contrasting scenes of logging and

² The preregistration is available at <https://osf.io/8fj3d/overview>.

³ Further documentation of the VR stimulus is provided in the Online Appendix (Section A.3).

deforestation. The experience culminated in a Google Earth-style globe visualisation presenting two conservation scenarios — protecting 20% or 50% of the Amazon — with the viewing scale automatically adjusting to show the geographical extent of each protection level. The presentation order of these scenarios was randomised across participants.

The VR condition was implemented within an environmentally controlled setting where humidity was set at 70%, temperature at 32°C, and a custom petrichor scent accord was used to recreate the smell of rain and wet soil. These multisensory elements were included to create a fully immersive and experientially coherent VR environment, allowing the study to examine immersive presentation as an integrated experience rather than isolating individual sensory components. While spatial visualisation is most directly relevant to conveying conservation scale, the additional sensory elements reflect the broader multisensory nature of immersive VR. Participants could navigate freely using head movements for 360° views, zoom using a joystick, and interact with information hotspots via gaze-based selection or controller pointing. Audio was delivered through the headset's integrated speakers. Unlike the computer-generated VR environments used in prior stated preference studies (Fang et al. 2021; Mokas et al. 2021), the present VR treatment combined 360° footage with multisensory environmental controls, reflecting contemporary immersive VR capabilities rather than visual immersion alone.

The video condition presented identical informational content in full high-definition 2D format on an iPad. A predetermined viewing path presented all hotspots at fixed timestamps, ensuring informational parity while restricting navigation to a fixed sequence. Video duration matched the nine-minute VR session with playback controls disabled. The video preserved all key environmental features and audio but omitted VR's climate controls and petrichor scent.

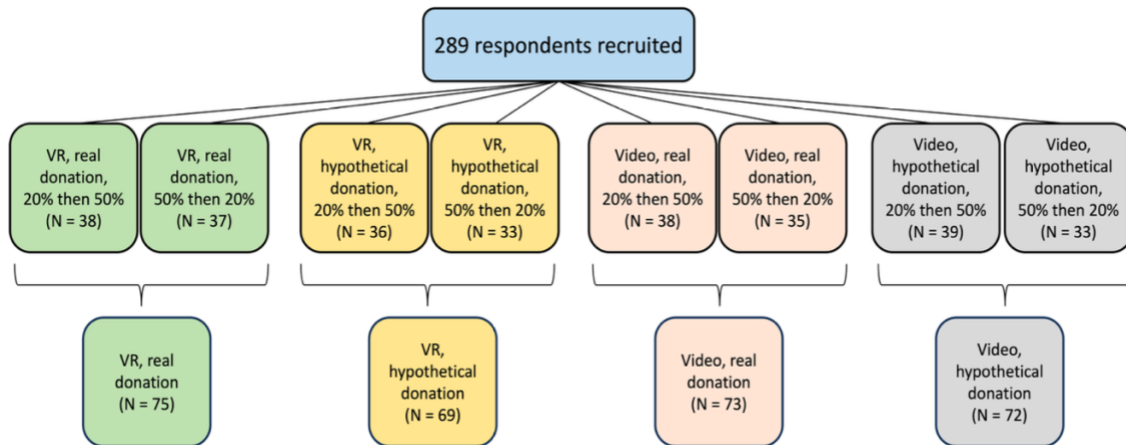
3.2. Experimental and Survey Design

The experiment used a 2×2 factorial design crossing presentation medium (VR vs. video) with donation type (real vs. hypothetical). The ordering of conservation programmes (20% vs. 50%) was randomised as a control measure rather than treated as an explicit factor, helping to account for potential ordering effects (Day et al. 2012; Halvorsen 1996; Clark and Friesen 2008).⁴

⁴ To test for potential ordering effects, a mixed-effects model with participant random intercepts included Order and the Scope × Order interaction. Neither the main effect of Order ($p = 0.481$) nor the interaction term ($p = 0.103$) was statistically significant.

Participants were independently randomised to presentation medium, donation condition, and scenario order, with session scheduling balanced across morning and afternoon slots to control for time-of-day effects (Nelson, DeVries, and Prendergast 2024). The resulting cell sizes varied minimally (Figure 1).

Figure 1 – Randomisation



The survey followed question sequencing consistent with established CV best practices (Johnston et al. 2017).⁵ After consent, all participants completed a two-minute non-environmental VR scenario of a virtual tour of the Alhambra in Spain, designed to standardise technological familiarity and mitigate novelty effects (Miguel-Alonso et al. 2023). An attitudinal section then assessed environmental concern, followed by knowledge and behavioural questions.

In the valuation section, participants experienced the VR or video presentation and then provided WTP values for both the 20% and 50% conservation scenarios. Those in the real donation condition were informed that one of their two bids would be randomly selected as binding for payment to the World Wide Fund for Nature (WWF).⁶ Upon completing the experiment, participants in this condition fulfilled their binding donation through a personalised link and received a WWF certificate confirming their contribution. Those in the hypothetical condition provided non-binding valuations. To maintain procedural equivalence, both conditions received a cheap talk script cautioning about potential overestimation and a budget reminder

⁵ The full survey is presented in the Online Appendix (Section A.1).

⁶ Donations were made directly to WWF-UK.

emphasising the implications for disposable income (Cummings and Taylor 1999). Participants also rated their certainty about their stated donation on a 1–10 scale, which was subsequently used for ex-post certainty-correction robustness checks. The session concluded with questions on motivations for WTP, VR experience and debriefing measures, and demographics. Strict protocols ensured experimental integrity, with the researcher maintaining neutral language throughout and following a standardised script across all sessions.⁷

3.3. Data Collection and Sample

Participants were recruited through the LSE Behavioural Research Lab from a panel of over 3,000 participants. Each received £10 in Amazon.co.uk gift vouchers as compensation and was entered into a lottery for a £25 Amazon.co.uk gift voucher. Separate experimental rooms were maintained for the VR and video conditions throughout data collection. To prevent unintended influence on expectations, the study was advertised using neutral language, avoiding priming terms such as “VR” or “virtual environment” (Orne 1962). Following a pilot study (N = 35) confirming technical functionality, the main data collection occurred from December 2024 to April 2025.

After data cleaning, the final analytical sample comprised 246 participants with 492 observations. Protest respondents (N = 26), identified as respondents reporting zero WTP values accompanied by justifications expressing organisational distrust rather than true zero valuations, were excluded (OECD 2018). Speedsters (N = 14), defined as responses completed in less than 17 minutes and 30 seconds — the minimum plausible engagement time given the fixed nine-minute stimulus — were excluded.⁸ Outliers (N = 3) above the 99th percentile (£100) were removed, a threshold consistent with inflation-adjusted estimates from a comparable Amazon conservation study (Horton et al. 2003). Although not representative of the general population, the predominantly university-based sample provides a controlled setting for testing the relative effects of presentation formats on valuation behaviour, which is the study’s primary objective.

⁷ The standardised script is presented in the Online Appendix (Section A.2).

⁸ This threshold was derived by summing minimum plausible engagement time: 30 seconds (consent), 9 minutes (fixed-duration media exposure), 3 minutes (VR pre-exposure), and 5 minutes (main survey).

3.4. Manipulation Checks

Three manipulation checks were used to verify that VR created a psychologically distinct experience from video. They measured dimensions identified as theoretically relevant in the VR literature (Slater 2009; Slater et al. 2020). Presence was assessed using “Feeling of being in the rainforest” rated on a scale of 0 (not at all) to 7 (extremely), capturing participants’ subjective sense of psychological immersion (Steuer 1992). Emotional engagement was measured through “To what extent did you feel emotionally moved by the rainforest experience?” (0 = not at all, 7 = extremely). Perceived realism was measured via “How do you find the overall survey? Is it realistic?” (0 = definitely not, 7 = definitely yes). This measure captures a broad assessment of the realism and credibility of the overall survey and valuation experience, rather than the realism of the virtual environment alone, a point considered when interpreting the exploratory analyses. These measures were subsequently used in exploratory analyses examining within-VR heterogeneity.

3.5. Empirical Strategy

Although scope sensitivity and hypothetical bias capture conceptually distinct validity dimensions, both analyses involve corner-solution donation behaviour and strongly right-skewed positive WTP values, motivating a common estimation framework. Given that true zero WTP responses comprised 26% of the data after excluding protest responses, all models employ a two-part hurdle framework (Cragg 1971).⁹ The first stage models the participation decision using a probit specification. The second stage models conditional WTP among positive responses using a generalised linear model (GLM) with Gamma distribution and log link to accommodate the right-skewed distribution of positive donations (Wooldridge 2010). This two-part structure allows different factors to influence participation and amount decisions, and has been adopted in CV research (Fosgerau and Bjørner 2006; Chu et al. 2020; Guo et al. 2023). An OLS specification with log-transformed WTP was estimated as a robustness check. The two stages are specified as follows:

$$P(WTP_{ij} > 0) = \Phi(\alpha_0 + \alpha_1 VR_i + \alpha_2 Scope_{ij} + \alpha_3 Donation_i + \alpha_4 Interaction_{it} + \delta' Z_i)$$

⁹ Refer to the Online Appendix (Section D.1) for visual and Shapiro–Wilk test diagnostics of distributional assumptions.

(Participation stage)

$$\log(E[WTP_{ij} | WTP_{ij} > 0]) = \beta_0 + \beta_1 VR_i + \beta_2 Scope_{ij} + \beta_3 Donation_i + \beta_4 Interaction_{it} + \Gamma' Z_i$$

(Conditional WTP stage)

Two specifications are estimated. In the scope sensitivity specification (Table 3, columns 1-2), the interaction term corresponds to $VR_i \times Scope_{ij}$, testing whether VR alters responsiveness to conservation scale. In the hypothetical bias specification (Table 3, columns 3-4), it corresponds to $VR_i \times Donation_i$, testing whether VR moderates the gap between hypothetical and real payment behaviour. Because the two specifications correspond to the preregistered tests of scope sensitivity and hypothetical bias, they are retained as the primary analyses. To examine additional interactions arising from jointly considering both validity dimensions, a fully interacted specification is reported in the Online Appendix (Table A34) and yields substantively similar results. VR_i indicates VR treatment (0 = video), $Scope_{ij}$ indicates the 50% conservation scenario (0 = 20%), $Donation_i$ indicates real payment (0 = hypothetical), and Z_i is a vector of controls including environmental concern, knowledge, behaviour, gender, education, and log income.

Bootstrap standard errors (2,000 replications) account for clustering of observations within participants. Hypotheses are tested using combined average marginal effects (AME) from the two-part model, integrating both participation and conditional amount decisions. Scope sensitivity is quantified using arc scope elasticity:

$$\varepsilon_{WTP,Q} = \frac{\Delta WTP}{\Delta Q} \times \frac{\bar{Q}}{\bar{WTP}}$$

where $\Delta Q = 0.30$, representing the change from 20% to 50% conservation, and the bar notation denotes midpoint values. Elasticity values between 0 and 1 indicate partial scope sensitivity, with empirical benchmarks ranging from 0.15 to 0.88 across environmental applications (Whitehead 2016; Kipperberg et al. 2024). Hypothetical bias is measured using calibration factors (CF), defined as the ratio of mean hypothetical to mean real WTP (Penn, Hu, and Ye 2024), where $CF > 1$ indicates hypothetical overstatement.

Exploratory within-VR analyses examine whether experiential features such as presence, emotion, and realism moderate scope sensitivity and hypothetical bias. Each moderator is

analysed in a separate specification to estimate its overall association with valuation behaviour. Presence, realism, and emotional engagement represent related but conceptually distinct dimensions of the immersive experience (Jung and Lindeman 2021) and are moderately correlated in the data ($r = 0.50\text{--}0.70$), reflecting their shared origins in the immersive experience. Separate specifications therefore estimate each dimension's association with valuation behaviour without conditioning on the other experiential dimensions, whereas a combined specification estimates only the unique partial association of each moderator conditional on the others. As the moderators capture overlapping aspects of immersion, the combined specification produces less precise interaction estimates, as reflected in the Online Appendix (Tables A19-A20).

4. RESULTS

4.1. Descriptive Statistics and Manipulation Check

The final analytical sample comprised 246 participants and 492 observations (two scope levels per participant). The sample was predominantly female (68.7%), young (52.9% aged 18–24), and highly educated, with 89.4% currently pursuing or having obtained at least a bachelor's degree. Balance tests indicated that treatment groups were generally balanced on observable characteristics, although the real donation group had higher education attainment and marginally higher environmental knowledge, and a higher proportion of females. These variables are included as controls in all specifications.¹⁰

Manipulation checks confirmed that VR created a distinct experience from video across all three experiential dimensions (Table 1). The largest difference emerged for presence, reflecting a shift from moderate to high immersion. Emotional engagement and perceived realism showed smaller but statistically significant increases. These results confirm that the experimental manipulation successfully altered participants' experiential responses, consistent with previous VR research (Mancuso et al. 2023; Gilpin, Gain, and Lipinska 2021).

Table 1: Manipulation Checks for VR Treatment

Variable	Video Mean	VR Mean	Diff.	p-value
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¹⁰ Full summary statistics and balance test tables are presented in the Online Appendix (Tables A2-A3).

Presence	3.778	5.264	1.486***	<0.001
Realism	4.906	5.248	0.342**	0.023
Emotion	4.350	4.736	0.386**	0.036
Participants	117	129	-	-

Notes: All variables measured on scales from 0 to 7. One-tailed tests. * $p < 0.1$; ** $p < 0.05$; *** $p < 0.01$.

Table 2 presents mean and median WTP by treatment cell. Both formats exhibited scope-sensitive valuations. Mean WTP increased by 67% for video and 77% for VR when conservation expanded from 20% to 50%, while median WTP increased from £3 to £5 in both formats. Real donations consistently produced lower estimates, with calibration factors ranging from 4.56 to 5.30, indicating substantial hypothetical bias.

Table 2: Willingness to Pay by Treatment Conditions (£)

Scope	Format	Hypothetical Donation (£)	Real Donation (£)	Mean (£)	Median (£)	CF
20%	Video	10.34 (1.47)	2.27 (0.37)	6.27 (0.84)	3.00	4.56
	VR	12.00 (1.41)	2.33 (0.44)	7.20 (0.85)	3.00	5.15
50%	Video	17.53 (2.37)	3.56 (0.66)	10.49 (1.38)	5.00	4.92
	VR	21.35 (2.56)	4.03 (0.73)	12.76 (1.54)	5.00	5.30

Notes: Standard errors in parentheses. CF = calibration factor (mean hypothetical / mean real WTP). WTP did not differ significantly between VR and video regardless of scope ($p = 0.276$; $p = 0.440$) or donation type ($p = 0.649$; $p = 0.193$).

4.2. Scope Sensitivity

Table 3 presents AMEs from two specifications, with columns (1) and (2) reporting scope sensitivity while (3) and (4) reporting hypothetical bias. At the population level, WTP increased by £4.61 for the larger conservation area, indicating scope-sensitive valuations across the full sample. The corresponding arc scope elasticity of 0.583 falls within the empirical range of 0.15–0.88 identified across environmental valuation studies (Whitehead 2016), indicating partial scope sensitivity.¹¹ Control variables generally exhibited expected relationships with participation and donation behaviour, broadly consistent with previous forest valuation studies (Horton et al. 2003; Mendonça and Tilton 2000; Ariyo, Okojie, and Ariyo 2018).

Table 3: Average Marginal Effects — Two-Part Hurdle Model

Scope Sensitivity	Hypothetical Bias
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¹¹ Scope elasticity = $(\Delta WTP / \Delta Q) \times (\bar{Q} / \bar{WTP}) = (4.61 / 0.30) \times (0.35 / 9.22) = 0.583$.

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Variables	(1) Probit	(2) Population	(3) Probit	(4) Population
<i>Scope_{ij}</i>	0.056*** (0.019)	4.610*** (0.673)	0.056*** (0.019)	4.705*** (0.706)
<i>VR_i</i>	0.050 (0.049)	1.450 (1.127)	0.051 (0.049)	1.808 (1.207)
<i>Donation_i</i>	-0.301*** (0.046)	-14.428*** (1.855)	-0.309*** (0.049)	-12.939*** (1.397)
<i>VR_i × Scope_{ij}</i>	0.002 (0.039)	0.866 (1.272)	—	—
<i>VR_i × Donation_i</i>	—	—	0.065 (0.100)	-2.284 (2.640)
<i>Env. Concern</i>	0.043** (0.020)	0.809 (0.574)	0.043** (0.020)	0.773 (0.562)
<i>Env. Knowledge</i>	0.046** (0.022)	0.903 (0.568)	0.046** (0.021)	0.924 (0.580)
<i>Env. Behaviour</i>	0.021 (0.027)	0.129 (0.835)	0.022 (0.027)	0.130 (0.832)
<i>Male</i>	-0.084 (0.056)	-0.938 (1.669)	-0.084 (0.055)	-1.017 (1.697)
<i>Education</i>	0.010 (0.034)	1.895** (0.837)	0.008 (0.035)	1.981** (0.827)
<i>Log Income</i>	0.058*** (0.019)	1.702*** (0.579)	0.059*** (0.019)	1.691*** (0.575)

Notes: Dependent variables: Probit models whether WTP>0; population-level models unconditional WTP in £. Bootstrapped standard errors in parentheses (2,000 replications); N=462 (30 observations excluded due to missing income data). Probit columns report AMEs on participation probability. Population columns report combined AMEs in £. — indicates variable not included in specification. Conditional AMEs of Scope: video = 4.149 ($p < 0.01$), VR = 5.014 ($p < 0.01$). Conditional AMEs of Donation: video = -11.722 ($p < 0.01$), VR = -14.006 ($p < 0.01$). * $p < 0.1$; ** $p < 0.05$; *** $p < 0.01$.

The conditional AMEs reveal that both formats achieved significant scope sensitivity, with video participants increasing WTP by £4.15 and VR participants by £5.01. The corresponding scope elasticities were 0.576 and 0.588, respectively.¹² Importantly, the interaction between VR and Scope was not statistically significant (£0.87), indicating that while both presentation formats

¹² These subgroup elasticities are calculated separately within each presentation format and therefore differ slightly from the pooled sample elasticity reported above. Scope elasticity = $(\Delta WTP / \Delta Q) \times (\bar{Q} / \bar{WTP}) = (4.15 / 0.30) \times (0.35 / 8.40) = 0.576$ for video and $(5.01 / 0.30) \times (0.35 / 9.94) = 0.588$ for VR.

elicited scope sensitivity, VR did not significantly amplify responsiveness to conservation scale relative to video.¹³

4.3. Hypothetical Bias

Columns (3) and (4) of Table 3 present results from the hypothetical bias specification. At the population level, participants donated £12.94 less when facing real financial consequences, reflecting substantial hypothetical bias. The overall calibration factor of 5.36 indicates that participants stated WTP approximately five times the actual donations, within the empirical range documented across valuation studies (Penn and Hu 2018).¹⁴ By presentation format, video participants exhibited a gap of £11.72 with a calibration factor of 5.13, while VR participants showed a slightly larger gap of £14.01 with a calibration factor of 5.62.¹⁵

The VR × Donation interaction was not statistically significant (−£2.28), indicating that VR did not reduce hypothetical bias relative to video. This result remained robust across certainty-correction thresholds (≥ 7 , ≥ 8 , and ≥ 9), with the VR × Donation interaction remaining non-significant under all specifications (see the Online Appendix Table A6).¹⁶ If immersive realism and psychological engagement strengthen economic consequentiality, one would expect a narrower gap between hypothetical and real contributions in the VR condition. Instead, VR altered the psychological experience of the valuation task without reducing hypothetical bias. However, because VR substantially altered participants' experiential responses, average treatment effects may conceal heterogeneity associated with experiential intensity within the VR condition. The following section therefore examines whether experiential dimensions are differentially associated with scope sensitivity and hypothetical bias within the VR condition.

4.4. Within-VR Heterogeneity

While average treatment effects did not differ significantly across formats, exploratory within-VR analyses reveal important heterogeneity. Tables 4 and 5 examine whether three experiential

¹³ Separate analyses within the VR condition indicated positive scope responsiveness under both hypothetical and real donation conditions, although the magnitude was substantially smaller under binding payment. Full results are reported in the Online Appendix (Table A16).

¹⁴ CF = Hypothetical/Real = 15.86/2.96 = 5.36.

¹⁵ CF = Hypothetical/Real = 14.56/2.84 = 5.13 (video); 17.04/3.03 = 5.62 (VR).

¹⁶ As a further test parallel to Mokas et al. (2021), stated certainty in donation amounts did not differ significantly between VR and video for either the 20% or 50% conservation scenario ($p = 0.444$ and $p = 0.269$, respectively).

dimensions — presence, emotion, and realism — moderate scope sensitivity and hypothetical bias, respectively, with each moderator analysed in a separate specification as described in Section 3.¹⁷ These dimensions capture complementary aspects of the immersive experience, with presence reflecting the sense of “being there”, emotion capturing the intensity of affective response, and realism reflecting the participants’ overall assessment of the credibility and realism of the valuation experience.

Table 4 presents the scope sensitivity moderator results. Presence and realism significantly moderated scope sensitivity within VR. Each additional unit of perceived realism was associated with a £1.33 increase in the WTP response to the larger conservation scope, and each unit of presence with a £1.03 increase. At the highest realism level (7), participants increased WTP by £8.02 for the larger conservation area, compared to a non-significant £1.38 at the lowest level (2). A similar gradient emerged for presence, ranging from –£0.07 at the lowest level to £7.14 at the highest. Emotion did not significantly moderate scope sensitivity.

Table 4: Within-VR Heterogeneity in Scope Sensitivity

	(1)	(2)	(3)
Variables	Presence	Emotion	Realism
<i>Scope_{ij}</i>	5.027*** (1.052)	4.971*** (1.027)	4.929*** (1.030)
<i>Donation_i</i>	–16.018*** (3.018)	–16.165*** (3.074)	–15.784*** (2.859)
<i>Experiential Dimension</i>	0.573 (0.750)	0.886 (0.637)	1.379* (0.767)
<i>Scope × Experiential Dimension</i>	1.030** (0.482)	0.569 (0.511)	1.328** (0.629)
<i>Controls</i>	Yes	Yes	Yes

Notes: Dependent variable is unconditional WTP in £, with table showing population-level AMEs within the VR condition. Bootstrapped standard errors in parentheses (2,000 replications); N=246 observations (12 observations excluded due to missing income data). Each column estimates a separate model interacting the named experiential dimension with conservation scope. “Experiential dimension” refers to the column-heading variable. Controls as in Table 3 with full specification in the Online Appendix (Table A24). Conditional AME of Scope at lowest/highest moderator values — Presence: –0.074 ($p > 0.05$) / 7.137; Emotion: 2.659 ($p > 0.05$) / 6.642 ($p < 0.01$); Realism: 1.380 ($p > 0.05$) / 8.019 ($p < 0.01$). * $p < 0.1$; ** $p < 0.05$; *** $p < 0.01$.

¹⁷ A combined specification is reported in the Online Appendix (Tables A19-A20).

Table 5 presents the hypothetical bias moderator results. In contrast to the scope sensitivity results, emotion significantly amplified hypothetical bias, with each additional unit of emotional intensity corresponding to a £2.85 increase in the gap between hypothetical and real donations. Perceived realism had a marginally significant effect (£2.58), while presence did not significantly moderate hypothetical bias.

Table 5: Within-VR Heterogeneity in Hypothetical Bias

Variables	(1) Presence	(2) Emotion	(3) Realism
<i>Scope_{ij}</i>	5.059*** (1.052)	5.159*** (1.081)	4.906*** (1.022)
<i>Donation_i</i>	-13.790*** (2.086)	-14.625*** (2.207)	-13.700*** (2.007)
<i>Experiential Dimension</i>	0.654 (0.873)	1.349* (0.703)	1.566* (0.950)
<i>Donation × Experiential Dimension</i>	-1.425 (1.496)	-2.845** (1.417)	-2.583* (1.492)
<i>Controls</i>	Yes	Yes	Yes

Notes: Dependent variable is unconditional WTP in £, with table showing population-level AMEs within the VR condition. Bootstrapped standard errors in parentheses (2,000 replications); N=246 (12 observations excluded due to missing income data). Each column estimates a separate model interacting the named experiential dimension with donation type. “Experiential dimension” refers to the column-heading variable. Controls as in Table 3 with full specification in the Online Appendix (Table A25). Conditional AME of Donation at lowest/highest moderator values — Presence: -6.583 ($p > 0.05$) / -16.559 ($p < 0.01$); Emotion: -2.730 ($p > 0.05$) / -22.645 ($p < 0.01$); Realism: -6.082 ($p > 0.05$) / -19.000 ($p < 0.01$). * $p < 0.1$; ** $p < 0.05$; *** $p < 0.01$.

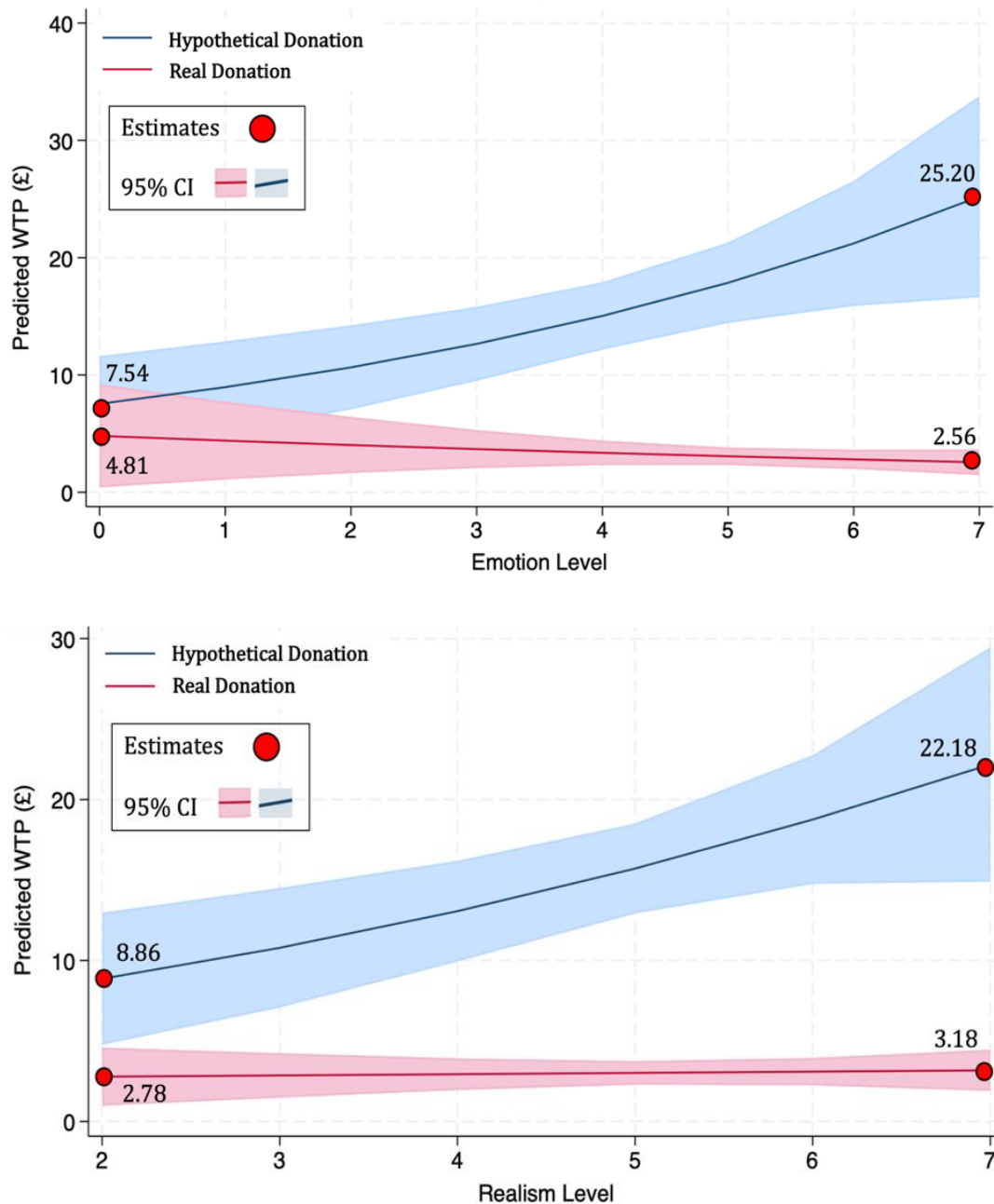
These results are consistent with an asymmetric affective amplification pattern, decomposed in Table 6. As emotional intensity increases from low (0) to high (7), hypothetical donations increase from £7.54 to £25.20, while real donations decrease from £4.81 to £2.56. The difference-in-differences estimate of £19.92 indicates that greater emotional engagement amplifies stated WTP without a corresponding increase in actual contributions. A similar pattern emerges for perceived realism, where hypothetical donations increase by £13.31 as realism rises, while real donations remain nearly constant (+£0.40), yielding a difference-in-differences estimate of £12.92. Figure 2 illustrates these diverging trajectories. Parallel within-video

analyses showed no significant moderation of hypothetical bias by any experiential dimension (see the Online Appendix Table A21).

Table 6: Within-VR Decomposition of Hypothetical and Real Donations (£)

	Hypothetical Donation (£)	Real Donation (£)	Hypothetical Bias (£)
Panel A: Emotion			
Low Emotion (0)	7.54** (3.06)	4.81 (6.06)	2.73 (6.97)
High Emotion (7)	25.20*** (5.18)	2.56*** (0.73)	22.65*** (5.13)
Change (High – Low)	17.66** (7.30)	–2.26 (6.50)	
Difference-in- Differences			19.92** (9.92)
Panel B: Realism			
Low Realism (2)	8.86*** (3.07)	2.78* (1.59)	6.08* (3.59)
High Realism (7)	22.18*** (5.00)	3.18*** (0.79)	19.00*** (4.90)
Change (High – Low)	13.31* (7.22)	0.40 (2.03)	
Difference-in- Differences			12.92* (7.46)

Notes: Bootstrapped standard errors in parentheses (2,000 replications). Hypothetical Bias = Hypothetical – Real donation. Difference-in-differences tests whether emotion/realism amplifies hypothetical bias. Conditional predictions are evaluated at the observed minimum and maximum values of each experiential dimension. Emotion ranged from 0–7, while the minimum observed realism score was 2. * $p < 0.1$; ** $p < 0.05$; *** $p < 0.01$.

Figure 2: Within-VR Predicted WTP by Experiential Intensity: Hypothetical vs Real Donations

Notes: No participants selected '0' or '1' for realism level. Predicted WTP from the two-part hurdle model within the VR condition. Hypothetical donations (blue) diverge from real donations (red) as experiential intensity increases. 95% confidence intervals shown.

A combined specification including all three experiential dimensions and their interactions, reported in the Online Appendix (Tables A19-A20), yields substantively similar conditional marginal effects, although individual interaction terms are attenuated by shared variance among the moderators. Additional exploratory analyses reported in the Online Appendix (Tables A17-A18) examined physiological discomfort and prior VR experience as potential moderators.

Higher discomfort reduced scope sensitivity, consistent with perceptual disruption impairing respondents' ability to differentiate conservation scales, but did not moderate hypothetical bias. Prior VR experience moderated neither dimension, providing no evidence that prior VR experience drove the observed patterns. These findings also contrast with Fang et al. (2021), where VR effects on hypothetical bias emerged only after excluding participants reporting simulator sickness.

Taken together, these findings suggest that different experiential dimensions influence distinct aspects of stated preference validity. Presence and perceived realism were associated with stronger scope differentiation, whereas emotional engagement was primarily associated with hypothetical overstatement. Perceived realism was the only experiential dimension linked to both validity criteria, strengthening scope sensitivity while simultaneously amplifying hypothetical bias.

4.5. Robustness Checks

The main findings were confirmed through a series of robustness checks reported in the Online Appendix (Section D). These include an OLS specification with log-transformed WTP, interval regression for hypothetical payment-card responses, functional-form and residual diagnostics, and sensitivity to the inclusion of outlier, speedsters, and protest respondents. All primary results, including the absence of a significant difference in scope sensitivity across formats, substantial hypothetical bias in both conditions, and the within-VR pattern of affective amplification, remained robust across specifications, suggesting that the multidimensional pattern of validity effects is not driven by modelling assumptions or sample exclusions.

5. DISCUSSION

5.1. Main Findings

This study provides the first assessment of how fully immersive VR affects both scope sensitivity and hypothetical bias in CV using a preregistered experiment. At the average treatment level, VR and video produced comparable scope-sensitive valuations with empirically plausible elasticities, and hypothetical bias did not differ significantly between formats. Both calibration

factors fell within the empirical range documented by Penn and Hu (2018), and this result remained robust to certainty corrections at multiple thresholds.

These aggregate effects conceal important heterogeneity within the VR condition. The three experiential dimensions were associated differently with scope sensitivity and hypothetical bias, suggesting that immersive presentation influences distinct dimensions of stated preference validity through different pathways. The following sections examine these patterns in greater detail.

5.2. Experiential Dimensions and Stated Preference Validity

5.2.1. Presence and Representational Processing

Presence was associated with stronger scope sensitivity, unlike emotional engagement, suggesting that scope differentiation within VR is more consistent with representational rather than affective channels. Participants reporting stronger feelings of “being there” showed stronger differentiation between conservation scales, a pattern consistent with spatial comprehension and scale visualisation as plausible cognitive pathways underlying scope sensitivity.

This interpretation aligns with the constructed preferences framework (Payne, Bettman, and Johnson 1992; Lichtenstein and Slovic 2006), in which respondents construct valuations during the elicitation process using available informational cues. Greater presence may strengthen this process by making conservation scale differences more concrete and perceptually accessible, helping respondents translate perceived differences in scale into correspondingly differentiated valuations.

The equivalence between VR and video at the average treatment level nevertheless suggests a task-specific threshold for scope differentiation. Although less immersive, the video condition presented the same environmental content and key spatial cues, and these appear sufficient for participants to distinguish meaningfully between protecting 20% and 50% of the Amazon rainforest. Once conservation scale differences become visually comprehensible, additional sensory immersion may not further strengthen scope discrimination. A secondary, more speculative explanation for the absence of an average VR advantage concerns breaks in presence. VR may create expectations of multisensory coherence that can be disrupted by

contradictory environmental cues, such as ambient room conditions or external sounds inconsistent with the virtual environment (Slater and Steed 2000). Such disruptions may attenuate the potential representational advantages of immersion for some participants, helping to explain why stronger presence was associated with scope sensitivity within VR without translating into a significant average advantage over video.

This association between presence and scope sensitivity extends prior research identifying presence as a pathway through which VR influences pro-environmental responses (Ahn, Bailenson, and Park 2014). Notably, however, presence did not significantly moderate hypothetical bias, distinguishing it from the other experiential dimensions examined below.

5.2.2. Emotional Engagement and Affective Amplification

Emotional engagement showed the opposite pattern. It did not significantly moderate scope sensitivity, consistent with evidence that emotionally engaging immersive content does not necessarily translate into stronger valuation responses or behavioural change (Martingano et al. 2023), but showed the strongest moderation pattern for the gap between hypothetical and real donations.¹⁸ As emotional intensity increased, hypothetical donations rose substantially while real donations remained relatively stable.

If immersive presentation strengthened economic consequentiality, one would expect a narrower gap between hypothetical and real contributions. Instead, the observed pattern suggests that emotional intensity amplified hypothetical valuation without proportionally affecting behaviour under real financial constraints. In hypothetical contexts where budget constraints are non-binding, participants reporting stronger emotional responses may have responded primarily to affective intensity and anticipated moral satisfaction rather than opportunity costs (Johansson-Stenman and Svedsäter 2012), consistent with warm glow effects (Kahneman and Knetsch 1992).

One interpretation is that actual financial commitment may require respondents to reconcile affective responses with tangible opportunity costs, activating more deliberative evaluation that

¹⁸ Martingano et al. (2023) randomly assigned 155 participants to 360° VR, VR with perspective-taking instructions, an audiobook control, or a VR control condition. Although VR increased short-run emotional empathy, it did not increase real charitable donations to UNICEF, and most empathic gains had dissipated after ten days.

tempers contributions (Rand, Greene, and Nowak 2012; Small, Loewenstein, and Slovic 2007). The relative stability of real donations across levels of emotional engagement is consistent with this account, although it may also partly reflect binding budget constraints within the predominantly student sample.

In addition, emotional engagement did not significantly moderate hypothetical bias in the video condition. One speculative explanation is that video maintains a more observational perspective, creating greater psychological separation between respondents and the conservation scenario (Khanna, Lee, and Kalantari 2025), whereas VR may increase the salience of emotional responses without proportionally affecting real financial behaviour. This pattern aligns with broader evidence that immersive media can increase empathy and donation intentions while producing inconsistent effects on actual giving (Kristofferson, Daniels, and Morales 2022; Walewijns, Heirman, and Daneels 2023). More generally, the findings suggest that technologies capable of intensifying emotional engagement may strengthen stated valuation responses without proportionally increasing real financial commitment.

5.2.3. Perceived Realism Across Both Validity Dimensions

Perceived realism differed from presence and emotional engagement in being associated with both validity outcomes. Higher perceived realism was associated with stronger scope differentiation, but was also marginally associated with a larger gap between hypothetical and real donations.

Unlike presence and emotional engagement, the realism measure asked participants how realistic they found the overall survey. It therefore captures a broader assessment of the credibility and plausibility of the valuation exercise, including the conservation scenario and the way in which it was presented, rather than the realism of the virtual environment alone. One possible interpretation is that respondents who regarded the survey and conservation scenario as more credible engaged more seriously with the valuation task. This may have encouraged closer attention to differences between conservation outcomes, contributing to stronger scope differentiation.

At the same time, greater credibility may have reinforced the sense that the proposed contribution related to a genuine conservation programme, increasing stated willingness to

contribute in the hypothetical condition. This is consistent with evidence that scenario credibility and perceived consequentiality are distinct, and that stronger credibility perceptions are associated with higher stated WTP even where information has little effect on consequentiality perceptions (Welling, Zawojka, and Sagebiel 2022). Perceived realism may therefore have strengthened belief in the scenario itself without changing the fact that the hypothetical donation carried no actual financial cost. This may help explain why hypothetical donations increased with perceived realism while real donations remained comparatively stable. However, this moderation was weaker than that observed for emotional engagement, and it was only marginally significant, with no corresponding significant moderation in the video condition.

5.3. Implications for Stated Preference Research

The central methodological implication is that presentation technologies may improve one dimension of stated preference validity while weakening another. Rather than uniformly improving valuation quality, immersive technologies appear to reshape the validity profile of stated preferences. The realism measure, which captures respondents' perceptions of overall survey and scenario realism, was the only experiential dimension associated with both dimensions of validity. Greater experiential realism should therefore not automatically be interpreted as evidence of superior elicitation quality.

For many environmental valuation applications, high-quality video may achieve scope-sensitive valuations comparable to VR at substantially lower cost and complexity. However, the experiential heterogeneity observed within VR suggests that immersive technologies may still offer important advantages in contexts where researchers are specifically interested in representational realism, psychological distance, or experiential mechanisms underlying preference formation. For researchers conducting large-scale or resource-constrained valuation studies, video may therefore represent a more practical approach capable of achieving scope-sensitive valuations without VR's equipment costs, technical requirements, and logistical constraints.

Similarly, for researchers who choose to employ VR, several design implications follow. First, immersive valuation environments should not be assumed to mitigate hypothetical bias. Substantial hypothetical bias persisted despite the use of a cheap-talk script and budget

reminder, and the main finding remained robust to ex-post certainty corrections. Immersive presentation should therefore not be regarded as a substitute for established bias-mitigation approaches. As presence was associated with stronger scope sensitivity without a corresponding increase in hypothetical bias, VR experiences designed to maximise spatial presence while moderating emotional intensity may offer a way to improve scope differentiation without this cost, although this hypothesis requires direct experimental testing.

Second, VR introduces experiential heterogeneity in hypothetical bias that standard calibration approaches may not adequately capture. For applied welfare analysis, standard calibration factors applied uniformly across respondents may inadequately adjust for variation in hypothetical bias arising from differences in experiential intensity. Collecting post-valuation experiential measures such as emotion, realism, and presence could inform the development of more targeted calibration approaches, although validated respondent-level methods remain a task for future research.

5.4. Limitations and Future Research

Several limitations warrant consideration. First, experiential dimensions such as emotion, realism, and presence were measured post-treatment and therefore cannot establish causal mediation claims. The within-VR heterogeneity analyses consequently identify associations between experiential intensity and valuation outcomes rather than causal pathways. Future studies that experimentally manipulate experiential features directly would provide stronger causal evidence.

Second, the study focused on a single environmental good characterised by high non-use values and considerable psychological distance. Whether affective amplification operates similarly for local, familiar, or primarily use-value environmental goods remains unclear.

Third, the predominantly student sample with relatively limited disposable income may have constrained real donations, contributing to the relative stability of actual payments across experiential intensity levels. Fourth, scope was operationalised as a single comparison between 20% and 50% conservation. While sufficient for testing scope sensitivity, this design does not characterise the full functional relationship between scope and willingness to pay.

Fifth, although the controlled experimental setting was necessary for delivering the VR treatment, the findings may not fully generalise to broader survey environments or large-scale field implementations. Finally, with approximately 60 participants per cell in the 2×2 design, the study may have had limited statistical power to detect moderate interaction effects. The non-significant interactions should therefore be interpreted alongside their confidence intervals rather than as definitive evidence of null effects.

Several directions for future research follow from these findings. Within-subject designs measuring both hypothetical and real willingness to pay for the same individuals would permit direct observation of individual-level bias and more precise moderation tests. Studies comparing environmental goods that differ in familiarity, psychological distance, and value type could help identify the boundary conditions of affective amplification. Factorial experiments combining VR with different bias-mitigation strategies could also test whether affective amplification can be mitigated through survey design. Finally, field experiments in real fundraising contexts would strengthen external validity and clarify whether the patterns identified here generalise beyond controlled experimental settings.

6. CONCLUSION

This study examined whether immersive VR improves two central dimensions of stated preference validity: scope sensitivity and hypothetical bias. Using a preregistered experiment comparing VR and high-quality video presentations for Amazon rainforest conservation, the findings show that immersive presentation does not uniformly improve valuation validity.

Both VR and video generated scope-sensitive valuations with empirically plausible elasticities, suggesting that clear visual communication of environmental scale may be sufficient for respondents to meaningfully differentiate between conservation outcomes for remote ecosystems with high non-use values. At the same time, substantial hypothetical bias persisted across both formats, and immersive presentation alone did not narrow the gap between stated and actual financial commitments.

The findings further reveal that experiential intensity within VR matters differently across validity criteria. Perceived realism and presence were associated with stronger scope sensitivity, whereas emotional engagement was more strongly associated with hypothetical overstatement.

Rather than functioning as a straightforward improvement in valuation quality, immersive technologies appear to reshape how environmental preferences are constructed and expressed. More broadly, immersive presentation should not be conflated with economic consequentiality. Technologies that make environmental scenarios feel more vivid, immersive, or emotionally engaging do not necessarily induce decision-making processes closer to those governing real financial behaviour.

These findings may also carry implications beyond stated preference methodology. Immersive and emotionally engaging media are increasingly used in conservation outreach with the expectation that heightened engagement will translate into stronger financial support. The present results suggest a more complex relationship in which technologies that intensify emotional engagement may strengthen stated support without proportionally increasing realised economic commitment. Future research should therefore focus not only on whether immersive technologies increase engagement, but on identifying which experiential features support differentiation between environmental outcomes, which amplify affective responding, and how these pathways interact with real economic commitment.

REFERENCES

- Ahn, Sun Joo (Grace), Jeremy N. Bailenson, and Dooyeon Park. 2014. Short- and long-term effects of embodied experiences in immersive virtual environments on environmental locus of control and behavior. *Computers in Human Behavior* 39 (October): 235-45.
- Ariyo, O. C., L. O. Okojie, and M. O. Ariyo. 2018. Villagers willingness to pay for forest conservation in Ibadan, Oyo State, Nigeria. *Asian Journal of Agricultural Extension, Economics & Sociology* 23 (4): 1-14.
- Azarby, Sahand, and Arthur Rice. 2022. Understanding the effects of virtual reality system usage on spatial perception: The potential impacts of immersive virtual reality on spatial design decisions. *Sustainability* 14 (16): 10326.
- Banerjee, Sanchayan, and Alipio Ferreira. 2024. Virtual reality is only mildly effective in improving forest conservation behaviors. *Scientific Reports* 14 (1): 28532.
- Bateman, Ian J., Brett H. Day, Andrew P. Jones, and Simon Jude. 2009. Reducing gain-loss asymmetry: A virtual reality choice experiment valuing land use change. *Journal of Environmental Economics and Management* 58 (1): 106-18.
- Birenboim, Amit, Martin Dijst, Dick Ettema, Joost de Kruijf, Geert de Leeuw, and Nico Dogterom. 2019. The utilization of immersive virtual environments for the investigation of environmental preferences. *Landscape and Urban Planning* 189 (September): 129-38.
- Breves, Priska, and Vivien Heber. 2020. Into the wild: The effects of 360° immersive nature videos on feelings of commitment to the environment. *Environmental Communication* 14 (3): 332-46.
- Brouwer, Roy, Rute Pinto, Anders Dugstad, and Ståle Navrud. 2022. The economic value of the Brazilian Amazon rainforest ecosystem services: A meta-analysis of the Brazilian literature. *PLOS ONE* 17 (5): e0268425.
- Burrows, James, Rebecca Newman, Jerry Genser, and Jeffrey Plewes. 2017. Do contingent valuation estimates of willingness to pay for non-use environmental goods pass the scope test with adequacy? A review of the evidence from empirical studies in the literature. In *Contingent Valuation of Environmental Goods: A Comprehensive Critique*, ed. Daniel McFadden and Kenneth Train, 82-152. Cheltenham, UK: Edward Elgar Publishing.
- Carson, Richard T. 1997. Contingent valuation: Theoretical advances and empirical tests since the NOAA panel. *American Journal of Agricultural Economics* 79 (5): 1501-7.
- Chang, Eunhee, Hyun Taek Kim, and Byounghyun Yoo. 2020. Virtual reality sickness: A review of causes and measurements. *International Journal of Human-Computer Interaction* 36 (17): 1658-82.
- Chopra, Vasudha, and Christian A. Vossler. 2026. Are we doing more harm than good? Using hypothetical bias reduction techniques in consequential decision settings. *Journal of the Association of Environmental and Resource Economists* 13 (4): 869-96.

- Chu, Xi, Jinyan Zhan, Chao Wang, Sultan Hameeda, and Xiaoru Wang. 2020. Households' willingness to accept improved ecosystem services and influencing factors: Application of contingent valuation method in Bashang Plateau, Hebei Province, China. *Journal of Environmental Management* 255 (February): 109925.
- Clark, Jeremy, and Lana Friesen. 2008. The causes of order effects in contingent valuation surveys: An experimental investigation. *Journal of Environmental Economics and Management* 56 (2): 195-206.
- Corso, Phaedra S., James K. Hammitt, and John D. Graham. 2001. Valuing mortality-risk reduction: Using visual aids to improve the validity of contingent valuation. *Journal of Risk and Uncertainty* 23 (2): 165-84.
- Cragg, John G. 1971. Some statistical models for limited dependent variables with application to the demand for durable goods. *Econometrica* 39 (5): 829-44.
- Cummings, Ronald G., and Laura O. Taylor. 1999. Unbiased value estimates for environmental goods: A cheap talk design for the contingent valuation method. *American Economic Review* 89 (3): 649-65.
- Day, Brett, Ian J. Bateman, Richard T. Carson, Diane Dupont, Jordan J. Louviere, Sanae Morimoto, Riccardo Scarpa, and Paul Wang. 2012. Ordering effects and choice set awareness in repeat-response stated preference studies. *Journal of Environmental Economics and Management* 63 (1): 73-91.
- Fang, Di, Rodolfo M. Nayga Jr., Grant H. West, Claudia Bazzani, Wei Yang, Benjamin C. Lok, Charles E. Levy, and Heather A. Snell. 2021. On the use of virtual reality in mitigating hypothetical bias in choice experiments. *American Journal of Agricultural Economics* 103 (1): 142-61.
- Flores, Bernardo M., Encarni Montoya, Boris Sakschewski, Nathália Nascimento, Arie Staal, Richard A. Betts, Carolina Levis, David M. Lapola, Adriane Esquivel-Muelbert, Catarina Jakovac, Carlos A. Nobre, Rafael S. Oliveira, Laura S. Borma, Da Nian, Niklas Boers, Susanna B. Hecht, Hans ter Steege, Julia Arieira, Isabella L. Lucas, Erika Berenguer, José A. Marengo, Luciana V. Gatti, Caio R. C. Mattos, and Marina Hirota. 2024. Critical transitions in the Amazon forest system. *Nature* 626 (7999): 555-64.
- Fosgerau, Mogens, and Thomas Bue Bjørner. 2006. Joint models for noise annoyance and willingness to pay for road noise reduction. *Transportation Research Part B: Methodological* 40 (2): 164-78.
- Gatti, Luciana V., Luana S. Basso, John B. Miller, Manuel Gloor, Lucas Gatti Domingues, Henrique L. G. Cassol, Graciela Tejada, Luiz E. O. C. Aragão, Carlos Nobre, Wouter Peters, Luciano Marani, Egidio Arai, Alber H. Sanches, Sergio M. Corrêa, Liana Anderson, Celso Von Randow, Caio S. C. Correia, Stephane P. Crispim, and Raiane A. L. Neves. 2021. Amazonia as a carbon source linked to deforestation and climate change. *Nature* 595 (7867): 388-93.
- Gilpin, Gina, James Gain, and Gosia Lipinska. 2021. The physiological signature of sadness: A comparison between text, film and virtual reality. *Brain and Cognition* 152 (August): 105734.

- Guo, Tongfang, Shuirong Wu, Xufeng Zhang, Chao Zhang, Jinrong Yang, and Shun Cheng. 2023. Measurement and influencing factors of willingness to accept payment for ecosystem service provision: A case study of a leading forest farm in China. *Forests* 14 (12): 2417.
- Haghani, Milad, Michiel C. J. Bliemer, John M. Rose, Harmen Oppewal, and Emily Lancsar. 2021a. Hypothetical bias in stated choice experiments: Part I. Macro-scale analysis of literature and integrative synthesis of empirical evidence from applied economics, experimental psychology and neuroimaging. *Journal of Choice Modelling* 41 (December): 100309.
- . 2021b. Hypothetical bias in stated choice experiments: Part II. Conceptualisation of external validity, sources and explanations of bias and effectiveness of mitigation methods. *Journal of Choice Modelling* 41 (December): 100322.
- Halvorsen, Bente. 1996. Ordering effects in contingent valuation surveys. *Environmental and Resource Economics* 8 (4): 485-99.
- Heberlein, Thomas A., Matthew A. Wilson, Richard C. Bishop, and Nora Cate Schaeffer. 2005. Rethinking the scope test as a criterion for validity in contingent valuation. *Journal of Environmental Economics and Management* 50 (1): 1-22.
- Horton, Bruce, Giordano Colarullo, Ian J. Bateman, and Carlos A. Peres. 2003. Evaluating non-user willingness to pay for a large-scale conservation programme in Amazonia: A UK/Italian contingent valuation study. *Environmental Conservation* 30 (2): 139-46.
- Johansson-Stenman, Olof, and Henrik Svedsäter. 2012. Self-image and valuation of moral goods: Stated versus actual willingness to pay. *Journal of Economic Behavior & Organization* 84 (3): 879-91.
- Johnston, Robert J., Kevin J. Boyle, Wiktor (Vic) Adamowicz, Jeff Bennett, Roy Brouwer, Trudy Ann Cameron, W. Michael Hanemann, Nick Hanley, Mandy Ryan, Riccardo Scarpa, Roger Tourangeau, and Christian A. Vossler. 2017. Contemporary guidance for stated preference studies. *Journal of the Association of Environmental and Resource Economists* 4 (2): 319-405.
- Jung, Sungchul, and Robert W. Lindeman. 2021. Perspective: Does realism improve presence in VR? Suggesting a model and metric for VR experience evaluation. *Frontiers in Virtual Reality* 2: 693327.
- Kahneman, Daniel. 2003. Maps of bounded rationality: Psychology for behavioral economics. *American Economic Review* 93 (5): 1449-75.
- Kahneman, Daniel, and Jack L. Knetsch. 1992. Valuing public goods: The purchase of moral satisfaction. *Journal of Environmental Economics and Management* 22 (1): 57-70.
- Kahneman, Daniel, Ilana Ritov, and David Schkade. 1999. Economic preferences or attitude expressions? An analysis of dollar responses to public issues. *Journal of Risk and Uncertainty* 19 (1-3): 203-35.
- Khanna, Jai, Anne Seo-young Lee, and Saleh Kalantari. 2025. The mediating role of technology affordances in virtual reality perspective-taking: A systematic literature review. *Virtual Reality* 29 (2): 66.

- Kipperberg, Gorm, Anders Dugstad, Ana Faria Lopes, Kristine Grimsrud, Henrik Lindhjem, and Ståle Navrud. 2024. Scope effects in stated preference research: From construct validity to economic significance. Unpublished manuscript, UiS Business School, University of Stavanger.
- Kristofferson, Kirk, Michelle E. Daniels, and Andrea C. Morales. 2022. Using virtual reality to increase charitable donations. *Marketing Letters* 33 (1): 75-87.
- Lee, Hui Min, and Benjamin J. Li. 2023. So far yet so near: Exploring the effects of immersion, presence, and psychological distance on empathy and prosocial behavior. *International Journal of Human-Computer Studies* 176: 103042.
- Lichtenstein, Sarah, and Paul Slovic, eds. 2006. *The Construction of Preference*. Cambridge: Cambridge University Press.
- Lopes, Ana Faria, and Gorm Kipperberg. 2020. Diagnosing insensitivity to scope in contingent valuation. *Environmental and Resource Economics* 77 (1): 191-216.
- Lovejoy, Thomas E., and Carlos Nobre. 2018. Amazon tipping point. *Science Advances* 4 (2): eaat2340.
- Mancuso, Valentina, Francesca Bruni, Chiara Stramba-Badiale, Giuseppe Riva, Pietro Cipresso, and Elisa Pedroli. 2023. How do emotions elicited in virtual reality affect our memory? A systematic review. *Computers in Human Behavior* 146 (September): 107812.
- Martingano, Alison Jane, Sara Konrath, Evan Henritze, and Adam D. Brown. 2023. The limited benefits of using virtual reality 360° videos to promote empathy and charitable giving. *Nonprofit and Voluntary Sector Quarterly* 52 (5): 1434-57.
- Mendonça, Augusto F., and John E. Tilton. 2000. A contingent valuation study of the environmental costs of mining in the Brazilian Amazon. *Minerals & Energy—Raw Materials Report* 15 (4): 21-32.
- Miguel-Alonso, Ines, Bruno Rodriguez-Garcia, David Checa, and Andres Bustillo. 2023. Countering the novelty effect: A tutorial for immersive virtual reality learning environments. *Applied Sciences* 13 (1): 593.
- Mokas, Ilias, Sebastien Lizin, Tom Brijs, Nele Witters, and Robert Malina. 2021. Can immersive virtual reality increase respondents' certainty in discrete choice experiments? A comparison with traditional presentation formats. *Journal of Environmental Economics and Management* 109 (September): 102509.
- Navrud, Ståle, and Jon Strand. 2018. Valuing global ecosystem services: What do European experts say? Applying the Delphi method to contingent valuation of the Amazon rainforest. *Environmental and Resource Economics* 70 (1): 249-69.
- Nelson, Katherine M., Eva Anggraini, and Achim Schlüter. 2020. Virtual reality as a tool for environmental conservation and fundraising. *PLOS ONE* 15 (4): e0223631.
- Nelson, Randy J., A. Courtney DeVries, and Brian J. Prendergast. 2024. Researchers need to better address time-of-day as a critical biological variable. *Proceedings of the National Academy of Sciences* 121 (30): e2316959121.

- OECD. 2018. *Cost-Benefit Analysis and the Environment: Further Developments and Policy Use*. Paris: OECD Publishing.
- Orne, Martin T. 1962. On the social psychology of the psychological experiment: With particular reference to demand characteristics and their implications. *American Psychologist* 17 (11): 776-83.
- Patterson, Zachary, Javad Mostofi Darbani, Ali Rezaei, John Zacharias, and Ali Yazdizadeh. 2017. Comparing text-only and virtual reality discrete choice experiments of neighbourhood choice. *Landscape and Urban Planning* 157 (January): 63-74.
- Payne, John W., James R. Bettman, and Eric J. Johnson. 1992. Behavioral decision research: A constructive processing perspective. *Annual Review of Psychology* 43 (1): 87-131.
- Penn, Jerrod M., and Wuyang Hu. 2018. Understanding hypothetical bias: An enhanced meta-analysis. *American Journal of Agricultural Economics* 100 (4): 1186-206.
- Penn, Jerrod M., Wuyang Hu, and Tao Ye. 2024. Efficacy of hypothetical bias mitigation techniques: A cross-country comparison. *Journal of Environmental Economics and Management* 125 (May): 102989.
- Rand, David G., Joshua D. Greene, and Martin A. Nowak. 2012. Spontaneous giving and calculated greed. *Nature* 489 (7416): 427-30.
- Rollins, Kimberly, and Audrey Lyke. 1998. The case for diminishing marginal existence values. *Journal of Environmental Economics and Management* 36 (3): 324-44.
- Schmidt, Jonas, and Tammo H. A. Bijmolt. 2020. Accurately measuring willingness to pay for consumer goods: A meta-analysis of the hypothetical bias. *Journal of the Academy of Marketing Science* 48 (3): 499-518.
- Shan, Jingzhu, Jingmei Li, Dandan Liu, Zhihua Xu, and Lewei Hong. 2023. Estimating the non-market value of island biodiversity in China: Controlling for warm glow bias in contingent valuation method. *Journal for Nature Conservation* 76 (December): 126509.
- Slater, Mel. 2009. Place illusion and plausibility can lead to realistic behaviour in immersive virtual environments. *Philosophical Transactions of the Royal Society of London. Series B, Biological Sciences* 364 (1535): 3549-57.
- Slater, Mel, and Anthony Steed. 2000. A virtual presence counter. *Presence: Teleoperators and Virtual Environments* 9 (5): 413-34.
- Slater, Mel, Cristina Gonzalez-Liencre, Patrick Haggard, Charlotte Vinkers, Rebecca Gregory-Clarke, Steve Jelley, Zillah Watson, Graham Breen, Raz Schwarz, William Steptoe, Dalila Szostak, Shivashankar Halan, Deborah Fox, and Jeremy Silver. 2020. The ethics of realism in virtual and augmented reality. *Frontiers in Virtual Reality* 1: 512499.
- Small, Deborah A., George Loewenstein, and Paul Slovic. 2007. Sympathy and callousness: The impact of deliberative thought on donations to identifiable and statistical victims. *Organizational Behavior and Human Decision Processes* 102 (2): 143-53.
- Steuer, Jonathan. 1992. Defining virtual reality: Dimensions determining telepresence. *Journal of Communication* 42 (4): 73-93.

- Sun, Rui, Yenchun Jim Wu, and Qian Cai. 2019. The effect of a virtual reality learning environment on learners' spatial ability. *Virtual Reality* 23 (4): 385-98.
- Tversky, Amos, and Daniel Kahneman. 1974. Judgment under uncertainty: Heuristics and biases. *Science* 185 (4157): 1124-31.
- Walewijns, David, Wannes Heirman, and Rowan Daneels. 2023. To give or not to give: Examining the prosocial effects of a 360° video endorsing a clean water charity. *Computers in Human Behavior* 145 (August): 107758.
- Welling, Malte, Ewa Zawojka, and Julian Sagebiel. 2022. Information, consequentiality and credibility in stated preference surveys: A choice experiment on climate adaptation. *Environmental and Resource Economics* 82 (1): 257-283.
- Whitehead, John C. 2016. Plausible responsiveness to scope in contingent valuation. *Ecological Economics* 128 (August): 17-22.
- Wooldridge, Jeffrey M. 2010. *Econometric Analysis of Cross Section and Panel Data*. 2nd ed. Cambridge, MA: MIT Press.
- Wunderling, Nico, Arie Staal, Boris Sakschewski, Marina Hirota, Obbe A. Tuinenburg, Jonathan F. Donges, Henrique M. J. Barbosa, and Ricarda Winkelmann. 2022. Recurrent droughts increase risk of cascading tipping events by outpacing adaptive capacities in the Amazon rainforest. *Proceedings of the National Academy of Sciences* 119 (32): e2120777119.
- Zawojka, Ewa, Anna Bartczak, and Mikołaj Czajkowski. 2019. Disentangling the effects of policy and payment consequentiality and risk attitudes on stated preferences. *Journal of Environmental Economics and Management* 93 (January): 63-84.
- Zhang, Chenyan, Aud Sissel Hoel, Andrew Perkis, and Saman Zadtootaghaj. 2018. How long is long enough to induce immersion? In *2018 Tenth International Conference on Quality of Multimedia Experience (QoMEX)*, 1-6. IEEE.
- Zhu, Wei, Shaoyu Guo, and Jinhua Zhao. 2021. Planning participants' preferential differences under immersive virtual reality and conventional representations: An experiment of street renewal. *Environment and Planning B: Urban Analytics and City Science* 48 (7): 1755-69.